E#6 - Module 6: In-Class Exercises #6

Turing Machine Design

Goal: Accept words with:

- **Odd number of a's**
- **Even number of b's**

State Definitions

I define four states to track the parity of a and b counts:

State	Meaning
I	Even a, Even b
J	Odd a, Even b
K	Even a, Odd b
L	Odd a, Odd b

Input alphabet (Σ): {a, b}

Tape alphabet (Γ): {a, b, ◻} (◻ = blank)

Head starts at the leftmost symbol and moves Right (R) on each step.

Any symbol not in {a, b} → REJECT

- **Start State**: I
- **Accept State**: J (Odd a, Even b)
- **Reject States**: All others

Transition Table

On blank (□) at end of input:

- If current state = J → ACCEPT

- If current state = I, K, or L → REJECT

Any other symbol → REJECT

Visual Representation of the Turing Machine

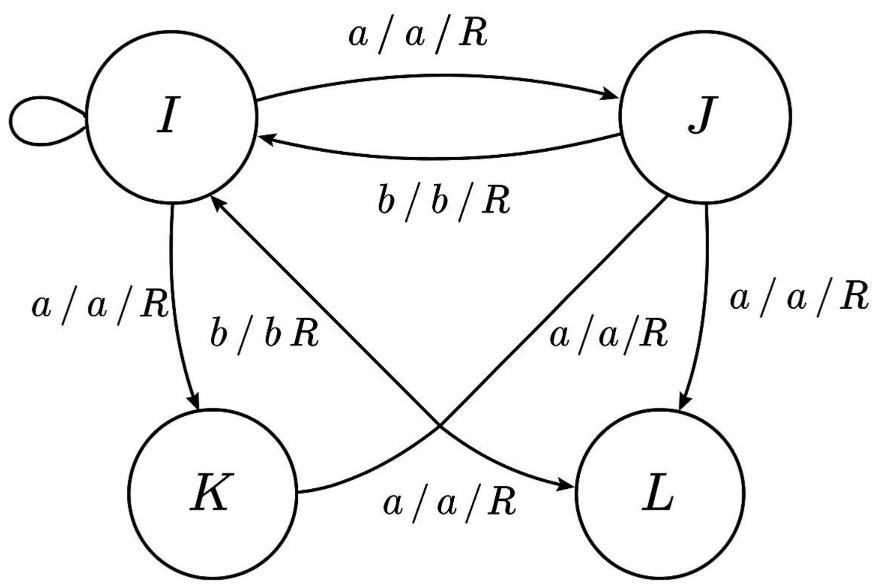

Figure 1: TM diagram for odd number of a's and even number of b's.

Current State	Read	Write	Move	Next State
I	a	a	R	J
I	b	b	R	K
J	a	a	R	I
J	b	b	R	L

Current State	Read	Write	Move	Next State
K	a	a	R	L
K	b	b	R	I
L	a	a	R	K
L	b	b	R	J

I | □ | □ | R | HALT (Reject)

J | □ | □ | R | HALT (Accept)

K | □ | □ | R | HALT (Reject)

L | □ | □ | R | HALT (Reject)

On blank (□) at end of input:

- If current state = J → ACCEPT

- If current state = I, K, or L → REJECT

Any other symbol → REJECT

Execution Examples

Below are the step-by-step executions for the required test words.

1 Word: ababa

- Step-by-step:
    - I → read a → J
    - J → read b → L
    - L → read a → K
    - K → read b → I

    - I → read a → J
- Final state: J **Accepted**

2 Word: abba

- Step-by-step:
    - I → read a → J
    - J → read b → L
    - L → read b → J
    - J → read a → I
- Final state: I **Rejected**

3 Word: bababa

- Step-by-step:
    - I → read b → K
    - K → read a → L
    - L → read b → J
    - J → read a → I
    - I → read b → K
    - K → read a → L
- Final state: L **Rejected**